JMMMU: A Japanese Massive Multi-discipline Multimodal Understanding Benchmark for Culture-aware Evaluation

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Abstract

Existing multimodal evaluation remains heavily English-centric, making it difficult to separate language transfer from culturally grounded understanding in non-English settings. We present JMMMU, a Japanese benchmark for expert-level multimodal evaluation with 1,320 questions and 1,118 images across 28 subjects. JMMMU is organized into two complementary parts: a culture-agnostic subset of 720 questions translated and localized from MMMU, enabling controlled English-to-Japanese comparison on matched content, and a culture-specific subset of 600 newly authored questions spanning Japanese Art, Japanese Heritage, Japanese History, and World History, targeting knowledge grounded in Japanese curricula and cultural context. The benchmark mixes multiple-choice and short open-ended questions, uses fixed answer-format prompts, and applies a deterministic parser with random fallback on unparsable multiple-choice outputs. Evaluating a diverse set of open and proprietary large multimodal models shows that performance remains low overall, with the best open-source model reaching 40.5% and GPT-4o reaching 58.6%. On the translated culture-agnostic subset, most models drop when moving from English to Japanese despite identical underlying content, while Japanese-specialized models reduce but do not eliminate this gap. On the culture-specific subset, models with similar translated-subset scores can diverge sharply, showing that Japanese language handling does not imply Japanese cultural competence. These results position JMMMU as a diagnostic benchmark for disentangling multilingual transfer from culture-aware multimodal understanding.

1 Introduction

Large multimodal models (LMMs) have improved rapidly, but the benchmarks used to measure them are still dominated by English data and English-centered cultural assumptions. Strong suites such as MMMU evaluate expert-level multimodal reasoning across many academic subjects, yet they do not tell us whether a model can transfer this competence into another language without changing the content, or whether it can handle culturally grounded content that cannot be reduced to translation alone (Yue et al., 2024a,c). This makes current evaluation incomplete for multilingual deployment.

The gap is especially visible in Japanese. Existing Japanese LMM benchmarks are useful, but they are small and mostly focused on commonsense or in-the-wild instruction-following settings rather than expert-level academic reasoning. As shown in Table 1, prior Japanese resources contain at most a few hundred questions, whereas JMMMU is both larger and more challenging. More importantly, previous Japanese evaluations typically do not disentangle two different failure modes: a model may underperform because it cannot robustly process Japanese, or because it lacks knowledge of Japanese culture and curriculum-specific content (SakanaAI, 2024b; Turing, 2024; SakanaAI, 2024a,c; Inoue et al., 2024).

JMMMU is designed to separate these factors. We build a *culture-agnostic* (CA) subset by translating and localizing 720 questions from the MMMU validation set while excluding subjects judged to be too culturally entangled for one-to-one transfer. This subset supports direct English-to-Japanese comparison under nearly matched content. We then add a *culture-specific* (CS) subset of 600 newly authored questions in four subjects aligned with Japanese educational practice: Japanese Art, Japanese Heritage, Japanese History, and World History. The resulting benchmark measures not just whether models "know Japanese," but whether they can reason about content that is culturally situated in Japan.

Empirically, JMMMU reveals three main findings. First, the benchmark remains difficult: the best open-source model, LLaVA-OV-7B, scores

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40.5% overall, while GPT-4o reaches 58.6%. Second, translation alone is not enough: across evaluated models, CA performance in Japanese is typically lower than CA performance in English, with several strong models losing more than 8 points on matched content. Third, CA and CS scores are correlated but not interchangeable: models with similar CA scores can differ substantially on CS, and GPT-4o outperforms Claude 3.5 Sonnet by 15.7 points on CS despite a much smaller 1.2-point gap on CA.

Our contributions are threefold. First, we introduce a Japanese expert-level multimodal benchmark with 1,320 questions and 1,118 images across 28 subjects. Second, we propose a benchmark design that explicitly separates translated, culture-agnostic evaluation from newly authored, culture-specific evaluation. Third, we provide a broad empirical study of open and proprietary LMMs, including English-to-Japanese comparisons, translation ablations, cross-subset analysis, and an error taxonomy for culture-specific failures.

2 Related Work

Expert-level multimodal benchmarks. MMMU established expert-level evaluation for multimodal reasoning across college subjects and remains a strong reference point for academic LMM assessment (Yue et al., 2024a). Subsequent work has strengthened benchmark robustness or broadened evaluation frameworks, for example by filtering out text-solvable questions in MMMU-Pro, emphasizing indispensable visual grounding in MMStar, or unifying evaluation pipelines in LMMS-EVAL (Yue et al., 2024c; Chen et al., 2024a; Zhang et al., 2024). JMMMU is complementary to these efforts. It does not attempt to replace English benchmarks or propose a new global evaluation framework; instead, it targets a missing measurement axis: expert-level multimodal evaluation in Japanese that can distinguish cross-lingual transfer from culture-aware competence.

Multilingual and multicultural multimodal evaluation. Prior multilingual VQA work has largely focused on translation-based extension of existing resources, including xGQA and MaXM, while more recent benchmarks such as MTVQA address multilingual text-centric visual reasoning and CVQA explicitly expands cultural coverage across many countries (Pfeiffer et al., 2021; Changpinyo et al., 2022; Tang et al., 2024; Romero et al., 2024). MaRVL similarly argues that culturally grounded multimodal reasoning cannot be captured by English-derived resources alone (Liu et al., 2021). JMMMU shares this motivation, but differs in two respects: it targets expert-level academic reasoning rather than mainly everyday VQA, and it includes a controlled translated subset so that language transfer and culture-specific reasoning can be analyzed separately within one benchmark.

Table 1: **Overview of Japanese LMM benchmarks.** JMMMU is the first benchmark that evaluates expert-level skills and is the largest among culture-aware benchmarks.

Benchmark	Culture	Level	Questions	Images
JA-VG-VQA-500 (SakanaAI, 2024b)	✗	Common sense	500	500
LLaVA-Bench-in-the-wild (Turing, 2024)	✗	Common sense	60	24
JA-Multi-Image-VQA (SakanaAI, 2024a)	✓	Common sense	55	39
JA-VLM-Bench-in-the-wild (SakanaAI, 2024c)	✓	Common sense	50	42
Heron Bench (Inoue et al., 2024)	✓	Common sense	102	21
JMMMU (Ours)	✓	Expert	1,320	1,118

Japanese multimodal benchmarks. Japanese evaluation resources for LMMs remain limited in scale and difficulty. Japanese Heron-Bench, JA-VLM-Bench-In-the-Wild, JA-Multi-Image-VQA, JA-VG-VQA-500, and translated LLaVA-Bench variants probe Japanese multimodal ability, but they primarily cover common-sense, instruction-following, or smaller-scale visual QA settings (Inoue et al., 2024; SakanaAI, 2024c,a,b; Turing, 2024). JMMMU extends this landscape toward expert-level evaluation, and Table 1 shows that it is substantially larger than prior culture-aware Japanese benchmarks.

3 Benchmark Design

3.1 Design Goals

JMMMU is designed around three goals. First, it should support *controlled cross-lingual comparison*: if a model drops from English to Japanese on the same underlying question, that degradation should be attributable primarily to language and localization rather than content change. Second, it should include *authentic Japanese cultural content* that cannot be reduced to translation from an English benchmark. Third, it should remain *expert-level and multimodal*, preserving the academic difficulty and heterogeneous image types that make MMMU challenging (Yue et al., 2024a).

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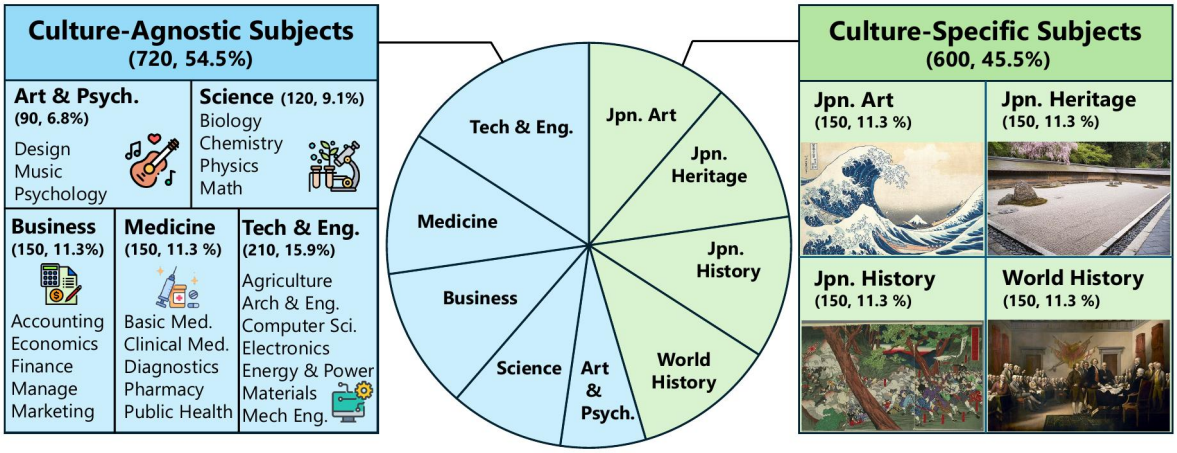
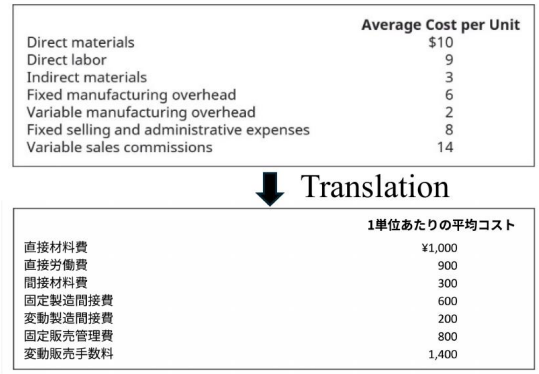


Figure 1: **Overview of JMMMU.** The benchmark contains 720 culture-agnostic questions translated from MMMU and 600 culture-specific questions written for Japanese cultural evaluation, for a total of 1,320 questions across 28 subjects.

3.2 Culture-Agnostic Subset

The CA subset starts from the MMMU validation set. Following the project overview and released benchmark assets, we exclude six subjects judged to be too culturally dependent for direct transfer: Art, Art Theory, Geography, History, Literature, and Sociology. The remaining 24 subjects yield 720 questions.

Each CA item is translated into Japanese and then refined by native Japanese annotators with subject familiarity. The refinement process aims to preserve answerability and difficulty while improving fluency, terminology, and style. Localization also extends to visuals when the image itself contains English text. In such cases, the English text is manually replaced with Japanese, and units or conventions are adapted only when required for natural Japanese presentation. For example, currency references are localized from dollars to yen without changing the underlying reasoning difficulty. Figure 2 shows one such image-localization example.



	Average Cost per Unit
Direct materials	$10
Direct labor	9
Indirect materials	3
Fixed manufacturing overhead	6
Variable manufacturing overhead	2
Fixed selling and administrative expenses	8
Variable sales commissions	14

Translation

	1単位あたりの平均コスト
直接材料費	¥1,000
直接労働費	900
間接材料費	300
固定製造間接費	600
変動製造間接費	200
固定販売管理費	800
変動販売手数料	1,400

Figure 2: **Example of image localization in the culture-agnostic subset.** English text embedded in the original figure is manually replaced with Japanese so that the translated benchmark tests Japanese multimodal understanding rather than English OCR.

3.3 Culture-Specific Subset

The CS subset contains 600 newly authored questions across four subjects: Japanese Art, Japanese Heritage, Japanese History, and World History, with 150 questions per subject. These subjects are intentionally aligned with topics that are salient in Japanese educational settings and public cultural knowledge, and they require more than everyday common sense.

The CS construction process uses redistributable images from Wikimedia Commons and authoring guidelines that emphasize concise problem statements, visually grounded reasoning, and plausible distractors. Questions are written to require either recognition of culturally situated visual evidence, access to Japan-specific historical or artistic knowledge, or both. Although World History is not Japan-specific in content, it is included in the CS slice because the questions are authored in Japanese and aligned with Japanese curriculum conventions rather than being inherited from English benchmark design.

3.4 Task Format and Evaluation Protocol

JMMMU mixes multiple-choice and short open-ended questions. Evaluation uses Japanese prompts

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Question: In <image 1>. $v_c = \sin 2\pi t$. Find an expression for i and calculate i at the instants $t = 0$.
(<image 1>において$v_c = sin\, 2\pi t$である。 iの式を求め、$t = 0$における iを計算しなさい。)

Option:
A. $3\pi\times10^{-5}$ A
B. $\pi\times10^{-5}$ A
C. $2\pi\times10^{-5}$ A
D. $\pi\times10^{-4}$ A

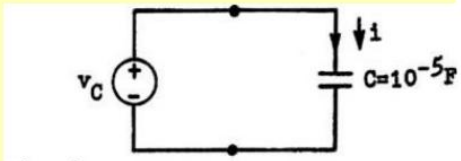

Fig. 1

Ground Truth: C. $2\pi\times10^{-5}$ A

GPT-4o's Answer in English:

B. $\pi\times10^{-5}$ A

GPT-4o's Answer in Japanese:
コンデンサの電流 i は、コンデンサの電圧 v_cの時間微分にキャパシタンス C を掛けたものです。すなわち、(The current i of a capacitor is the capacitance C multiplied by the time derivative of the capacitor's voltage v_c. That is,) $i = C\frac{dv_c}{dt}$
与えられた$v_c = \sin 2\pi t$を微分します。(Given $v_c = \sin 2\pi t$, we differentiate:) $\frac{dv_c}{dt} = \frac{d}{dt}(\sin 2\pi t) = 2\pi\cos(2\pi t)$
したがって、(Thus,) $i = C \cdot 2\pi\cos(2\pi t)$
ここで、$C = 10^{-5}$Fですので、(Here, $C = 10^{-5}$F, so)$i = 10^{-5} \cdot 2\pi\cos(2\pi t)$ $t = 0$のとき、(At $t = 0$,) $i = 10^{-5} \cdot 2\pi\cos(0)$
$\cos(0) = 1$なので、(Since $\cos(0) = 1$,) $i = 10^{-5} \cdot 2\pi \cdot 1 = 2\pi \cdot 10^{-5}$A
したがって正しい選択肢は**C**です。(Therefore, the correct choice is **C**.)

Figure 3: **Matched item before and after localization.** The localized version preserves the reasoning target while removing incidental English text, making CA gaps easier to interpret as Japanese transfer errors.

that require a constrained answer format: a single option letter for multiple-choice questions and a short phrase for open-ended questions. We use a rule-based parser to extract final answers from model outputs. For multiple-choice items, when the output is unparsable, the evaluation assigns a random option rather than dropping the item. This policy makes the benchmark robust to verbose or format-violating generations while keeping the scoring fully reproducible.

The benchmark is intended as an evaluation resource rather than a training set or a single scalar leaderboard. CA scores should be interpreted primarily as a measure of Japanese transfer under matched content, while CS scores probe culture-aware competence on newly authored Japanese material. The overall score is useful as a summary, but the diagnostic value of JMMMU comes from analyzing these two subsets separately.

4 Benchmark Validation

Coverage and diversity. Figure 1 summarizes the composition of the benchmark. The 720 CA questions cover five broad discipline groups inherited from MMMU: Art and Psychology, Business, Science, Health and Medicine, and Technology and Engineering. The 600 CS questions add four subjects that are absent from culture-agnostic transfer settings. Overall, JMMMU contains 1,118 images for 1,320 questions, preserving substantial visual heterogeneity.

Scale relative to prior Japanese resources. Table 1 shows that JMMMU is substantially larger than prior Japanese benchmarks we compare against and, unlike those resources, explicitly targets expert-level reasoning rather than commonsense evaluation. This matters because evaluation conclusions drawn from small, everyday-question datasets do not necessarily transfer to academically demanding multimodal tasks.

Quality control. The benchmark construction process involves 19 university students, including the authors, all native in Japanese and assigned to translation refinement or culture-specific authoring according to subject familiarity. Quality control focuses on linguistic naturalness, domain terminology, cultural correctness, and distractor plausibility. We view these checks as necessary for benchmark usability, especially because literal translation can

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create awkward or misleading Japanese prompts. For CA, reviewers also verify that translation and image localization preserve the original answer key and do not introduce new shortcuts. For CS, reviewers reject items that can be solved from generic commonsense alone without attending to Japan-specific evidence in the image or question.

The model results also serve as a benchmark-level sanity check. Random performance is well below strong-model accuracy, yet even GPT-4o remains far from ceiling, indicating that the benchmark is neither trivial nor saturated. At the same time, CA preserves enough shared structure to expose measurable English–Japanese transfer gaps, while CS changes several model rankings among otherwise similar systems. This pattern supports the claim that JMMMU is both comparable and diagnostically informative.

Known limitations. JMMMU does not claim exhaustive coverage of Japanese culture or of multimodal reasoning in Japanese. The CS subset is concentrated in history, heritage, and art-related areas, which is appropriate for culture-aware probing but does not cover all domains where culture may matter. In addition, the released materials do not report inter-annotator agreement, so reliability is supported by expert review and downstream model behavior rather than formal agreement statistics. Finally, as with other academic QA benchmarks, some items remain partially solvable from text alone; we quantify this explicitly in Section 5.2 instead of treating multimodal score as pure visual competence.

5 Benchmark Results

5.1 Experimental Setup

We evaluate a broad set of representative open and proprietary LMMs, including LLaVA variants (Liu et al., 2023, 2024; Li et al., 2024), InternVL (Chen et al., 2023, 2024b), CogVLM2 (Hong et al., 2024), xGen-MM (Xue et al., 2024), Phi-3 Vision models (Abdin et al., 2024), Pangea (Yue et al., 2024b), Gemini 1.5 Pro (Reid et al., 2024; DeepMind, 2024), Claude 3.5 Sonnet (Anthropic, 2024), and GPT-4o (OpenAI, 2024). We also report random-choice performance and a text-only GPT-4o baseline to measure how much signal can be extracted without images.

5.2 Overall Results

Table 2 reports the main results. JMMMU is challenging across the board. The best open-source model, LLaVA-OV-7B, reaches 40.5% overall, while the best proprietary model, GPT-4o, reaches 58.6%. Both figures are far from ceiling, especially given that random guessing already achieves 24.8% because many questions are multiple-choice.

The benchmark is also strongly diagnostic across subsets. On CA, the best open-source score is 38.5% and GPT-4o reaches 51.8%. On CS, the best open-source score rises to 47.0% with Pangea-7B, while GPT-4o reaches 66.7%. The CS slice is therefore not uniformly harder than CA, but it is more discriminative among strong models. For example, GPT-4o exceeds Claude 3.5 Sonnet by only 1.2 points on CA but by 15.7 points on CS, indicating that matched-content Japanese transfer and culture-aware competence are distinct evaluation targets.

Subject-level scores reveal additional structure. Japanese Heritage is the hardest CS subject for open models: the best open-source score is 40.0% from EvoVLM JP v2, versus 70.7% for GPT-4o. By contrast, World History is comparatively easier for strong models, with LLaVA-OV-7B reaching 68.0% and GPT-4o reaching 76.7%.

The text-only baseline is also informative. GPT-4o without images still achieves 38.1% overall, including 40.3% on CA and 35.5% on CS. This confirms that JMMMU, like other academic QA benchmarks, contains a nontrivial amount of text-accessible or prior-knowledge signal. At the same time, vision remains important: adding images improves GPT-4o by 20.5 points overall, 11.5 points on CA, and 31.2 points on CS.

5.3 Effect of Translation and Localization

Table 3 isolates the effect of translating text and localizing images on a subset of models. Performance often drops when English input is converted into Japanese. For instance, LLaVA-OV-7B falls from 45.0% with English text and images to 38.3% when only the text is translated, and to 35.6% when both text and images are localized. InternVL2-8B and Phi-3.5v show similar degradation. This pattern indicates that Japanese evaluation cannot be approximated by simply reusing English benchmarks with translated prompts.

The degradation is not universal. Japanese-oriented models such as LLaVA-CALM2 and

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Table 2: **Overall results.** CA (EN) shows the result on culture agnostic subset in English. The rest of the results are average and individual subjects' scores on JMMMU. †denotes Japanese LMMs. The best-performing model among open source and proprietary models are **in bold**. Overall, the performance is up to 40.5% for open-source, and 58.6% for proprietary models, leaving great room for improvement.

Models	Overall (1,320)	CS (600)	CA (720)	CA (EN) (720)	Jpn. Art (150)	Jpn. Heritage (150)	Jpn. History (150)	World History (150)	Art & Psych. (90)	Business (150)	Science (120)	Health & Medicine (150)	Tech & Eng. (210)
Random	24.8	25.0	24.6	24.6	25.0	25.0	25.0	25.0	25.4	25.0	22.8	25.6	24.3
Open Source													
LLaVA-OV-0.5B	26.0	23.3	28.2	29.4	22.7	22.7	24.0	24.0	26.7	27.3	24.2	30.7	30.0
InternVL2-2B	28.3	29.2	27.6	31.9	31.3	22.7	30.7	32.0	30.0	30.0	30.8	25.3	24.8
xGen-MM	28.6	28.2	28.9	35.7	30.0	20.7	22.7	39.3	32.2	21.3	22.5	36.7	31.0
Phi-3v	29.5	26.5	31.9	37.6	31.3	18.7	29.3	26.7	26.7	28.7	25.8	37.3	36.2
LLaVA-1.6-13B	31.1	33.7	29.0	29.9	32.0	24.0	32.0	46.7	25.6	28.7	30.0	34.0	26.7
Idefics2-8B	31.9	37.0	27.6	35.1	40.7	24.0	30.0	53.3	32.2	22.7	22.5	32.0	29.0
Phi-3.5v	32.4	34.3	30.8	39.2	37.3	27.3	35.3	37.3	27.8	31.3	30.0	36.7	28.1
†LLaVA CALM2	34.9	41.5	29.4	29.9	42.7	36.7	40.0	46.7	27.8	26.0	26.7	34.0	31.0
Mantis 8B	35.5	39.5	32.2	36.0	42.0	30.0	35.3	50.7	37.8	28.0	31.7	37.3	29.5
CogVLM2-19B	36.1	39.7	33.1	36.8	39.3	24.0	36.0	59.3	28.9	32.7	30.8	30.0	38.6
Idefics3-8B	37.3	42.8	32.8	36.9	43.3	24.7	**42.0**	61.3	34.4	28.0	26.7	38.0	35.2
†EvoVLM JP v2	38.1	45.2	32.2	33.9	44.0	**40.0**	**42.0**	54.7	32.2	28.7	28.3	**38.7**	32.4
InternVL2-8B	38.3	42.5	34.7	43.3	41.3	38.0	35.3	55.3	40.0	36.0	**34.2**	34.0	32.4
Pangea-7B	39.7	**47.0**	33.6	35.9	**46.0**	35.3	40.0	66.7	40.0	30.0	31.7	43.3	27.6
LLaVA-1.6-34B	39.8	43.2	37.1	45.7	42.0	36.0	40.7	54.0	**42.2**	**41.3**	25.0	36.7	39.0
LLaVA-OV-7B	**40.5**	43.0	**38.5**	45.1	36.0	30.7	37.3	**68.0**	41.1	36.7	31.7	**38.7**	**42.4**
Proprietary													
Claude 3.5 Sonnet	50.8	51.0	50.6	52.1	39.3	46.7	54.7	63.3	**53.3**	**56.7**	**51.7**	55.3	41.0
Gemini 1.5 Pro	51.5	60.3	44.2	51.1	54.7	55.3	55.3	76.0	51.1	44.0	44.2	48.0	38.6
GPT-4o	**58.6**	**66.7**	**51.8**	52.1	**60.7**	**70.7**	**58.7**	**76.7**	**53.3**	55.3	45.8	**61.3**	**45.2**
Text Only													
GPT-4o text	38.1	35.5	40.3	44.9	32.7	32.0	35.3	42.0	38.9	36.0	41.7	45.3	39.5

Table 3: **The effect of translation.** Each column shows the model performance when image (I) and text (T) are in Japanese (jp) or in English (en). Δ_i shows the difference from $I_{en}T_{en}$.

Model	$I_{en}T_{en}$	$I_{en}T_{jp}(\Delta_1)$	$I_{jp}T_{jp}(\Delta_2)$
LLaVA-1.6-13B	26.4	31.9 (+5.5)	29.2 (+2.8)
Phi-3.5v	39.2	33.6 (-5.6)	31.1 (-8.1)
LLaVA-CALM2	29.4	28.3 (-1.1)	31.4 (+2.0)
CogVLM2-19B	32.8	31.9 (-0.9)	34.4 (+1.6)
EvoVLM JP v2	30.0	30.8 (+0.8)	28.6 (-1.4)
InternVL2-8B	43.9	38.3 (-5.6)	37.2 (-6.7)
LLaVA-1.6-34B	43.6	40.8 (-2.8)	38.9 (-4.7)
LLaVA-OV-7B	45.0	38.3 (-6.7)	35.6 (-9.4)

EvoVLM JP v2 are much less sensitive, and LLaVA-CALM2 slightly improves after full localization. These cases support the intended interpretation of the CA subset: it is useful for measuring how well multilingual or Japanese-specialized models transfer expert-level multimodal reasoning into Japanese when the underlying content is held approximately constant.

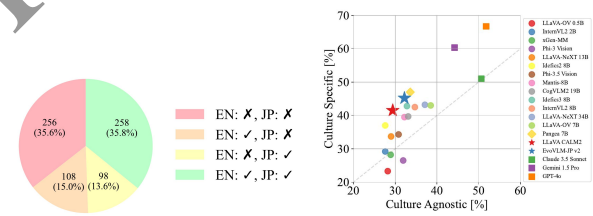

Figure 4: **Diagnostic analyses enabled by JMMMU.** Left: GPT-4o consistency across the English and Japanese versions of the 720 culture-agnostic questions. Right: relationship between culture-agnostic and culture-specific scores across evaluated models.

6 Analysis

6.1 Language Consistency on Matched Content

Figure 4 (left) examines GPT-4o on the 720 matched CA questions in English and Japanese. The model answers 258 questions correctly in both languages and 256 incorrectly in both, but 206 questions (28.6%) change correctness depending on the language: 108 are correct only in English, and 98 only in Japanese. This shows that even the strongest model is not simply preserving its

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decisions across translation. Language transfer introduces substantial instability at the item level, which would be invisible from aggregate accuracy alone.

This instability complements the average score gaps in Table 2. Across the reported multimodal models, CA performance in English exceeds CA performance in Japanese by 4.4 points on average, with gaps as large as 8.6 points for InternVL2-8B and LLaVA-1.6-34B. Japanese-specialized models narrow this gap: LLaVA-CALM2 loses only 0.5 points and EvoVLM JP v2 loses 1.7 points. JMMMU therefore exposes both model-level and question-level multilingual brittleness.

6.2 Culture-Agnostic and Culture-Specific Scores Are Related but Distinct

Figure 4 (right) plots CS scores against CA scores. The two measures are positively correlated, but the spread above the diagonal is large. Most strong models perform better on CS than on CA, yet the magnitude of this advantage varies substantially. Among open models, LLaVA-CALM2 and EvoVLM JP v2 stand out with much stronger CS than CA performance, suggesting that Japanese specialization helps more on culturally grounded questions than on translated academic content. Among proprietary models, GPT-4o and Gemini 1.5 Pro achieve notably high CS scores relative to their CA scores.

The main implication is that CA cannot serve as a proxy for CS. A model can transfer English-origin questions into Japanese reasonably well and still underperform on authentic Japanese content. Conversely, a Japanese-oriented model may gain more from culture-specific knowledge or training than from translated expert QA. This is precisely the distinction JMMMU is built to measure.

6.3 Error Taxonomy on Culture-Specific Questions

Figure 5 summarizes the benchmark's culture-specific error analysis. The largest category is *lack of knowledge* (53.8%), followed by *image recognition errors* (30.8%), *answer rejection* (10.6%), and *textual misunderstanding* (4.8%). This distribution is useful for interpreting what low CS performance means. The dominant issue is not merely inability to parse Japanese text; instead, models frequently fail because they do not know the underlying cultural or historical facts, or because they cannot connect those facts to the relevant visual evidence.

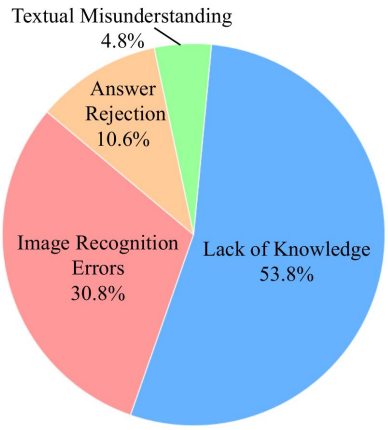

Figure 5: **Error distribution on culture-specific questions.** Lack of knowledge is the dominant failure mode, followed by image-recognition errors.

The nontrivial share of image-recognition errors also matters. JMMMU is not a pure knowledge quiz translated into Japanese. Many failures arise from visual perception mistakes, especially on artworks, heritage sites, or historically dense imagery where correct recognition is a prerequisite for retrieving the relevant knowledge. This reinforces the benchmark's role as a multimodal rather than text-only cultural evaluation.

7 Discussion

JMMMU is useful because it measures something that existing benchmarks tend to conflate. The CA subset measures Japanese transfer on matched expert-level multimodal content. The CS subset measures culturally grounded multimodal competence on newly authored Japanese material. Reporting only one of these would hide important differences in model behavior.

At the same time, the benchmark has clear limits. First, the CS subset covers only four subjects, so JMMMU should not be interpreted as a complete evaluation of Japanese culture. Second, some CA items remain partially solvable from text alone, as shown by the text-only GPT-4o baseline. Third, the construction process relies on expert review rather than formal inter-annotator agreement statistics. These limitations do not negate the benchmark's value, but they do constrain the conclusions that should be drawn from it.

We therefore view JMMMU as a diagnostic benchmark rather than a single leaderboard. It is most informative when used to compare English

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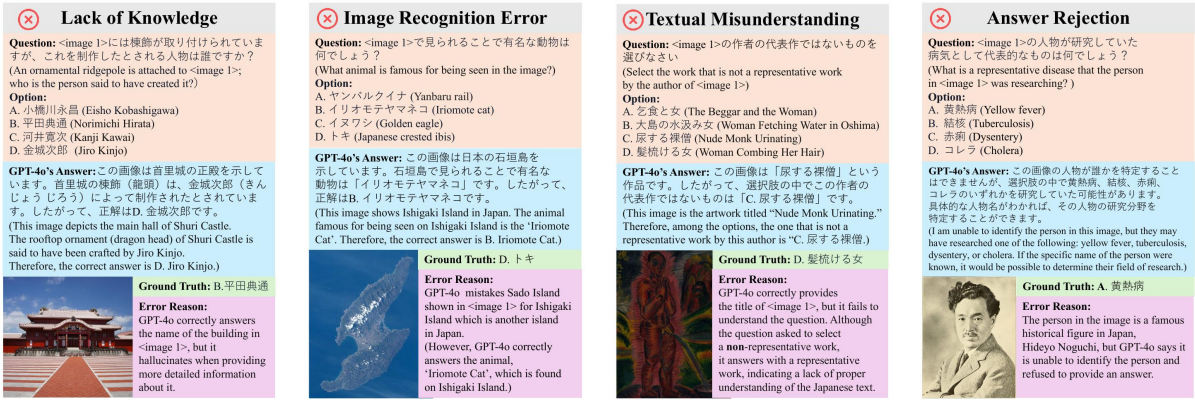

Figure 6: **Representative CS failures.** Left to right: lack of knowledge, image recognition error, textual misunderstanding, and answer rejection. Poor CS performance reflects several failure modes, not one artifact.

and Japanese performance on CA, to analyze ranking changes between CA and CS, and to inspect failure types on culture-specific items. Under this usage, it exposes a robust empirical pattern: strong Japanese language handling does not imply strong Japanese cultural understanding. In practice, we recommend reporting CA, CS, and CA(EN) together rather than only the overall score. This triplet separates cross-lingual transfer loss from culture-specific competence and makes it easier to detect models whose aggregate score hides opposite trends across the two subsets.

8 Conclusion

We introduced JMMMU, a Japanese expert-level multimodal benchmark with 1,320 questions and 1,118 images across 28 subjects. Its central design choice is to separate translated culture-agnostic evaluation from newly authored culture-specific evaluation, allowing cleaner diagnosis of language transfer versus cultural competence. Experiments across a broad range of LMMs show that current systems remain far from saturated, often lose accuracy when moving from English to Japanese even on matched content, and can diverge sharply on Japanese culture-specific questions despite similar translated-subset scores. JMMMU therefore provides a more informative basis for evaluating multilingual multimodal systems in Japanese and for studying how language ability and cultural knowledge interact in LMM performance.

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References

Marah Abdin, Sam Ade Jacobs, Ammar Ahmad Awan, Jyoti Aneja, Ahmed Awadallah, Hany Awadalla, Nguyen Bach, Amit Bahree, Arash Bakhtiari, Harkirat Behl, et al. 2024. Phi-3 technical report: A highly capable language model locally on your phone. *arXiv preprint arXiv:2404.14219*.

Anthropic. 2024. Claude 3.5 sonnet. https://www.anthropic.com/news/claude-3-5-sonnet.

Soravit Changpinyo, Linting Xue, Michal Yarom, Ashish V Thapliyal, Idan Szpektor, Julien Amelot, Xi Chen, and Radu Soricut. 2022. Maxm: Towards multilingual visual question answering. *arXiv preprint arXiv:2209.05401*.

Lin Chen, Jinsong Li, Xiaoyi Dong, Pan Zhang, Yuhang Zang, Zehui Chen, Haodong Duan, Jiaqi Wang, Yu Qiao, Dahua Lin, et al. 2024a. Are we on the right way for evaluating large vision-language models? In *NeurIPS*.

Zhe Chen, Weiyun Wang, Hao Tian, Shenglong Ye, Zhangwei Gao, Erfei Cui, Wenwen Tong, Kongzhi Hu, Jiapeng Luo, Zheng Ma, et al. 2024b. How far are we to gpt-4v? closing the gap to commercial multimodal models with open-source suites. *arXiv preprint arXiv:2404.16821*.

Zhe Chen, Jiannan Wu, Wenhai Wang, Weijie Su, Guo Chen, Sen Xing, Muyan Zhong, Qinglong Zhang, Xizhou Zhu, Lewei Lu, Bin Li, Ping Luo, Tong Lu, Yu Qiao, and Jifeng Dai. 2023. Internvl: Scaling up vision foundation models and aligning for generic visual-linguistic tasks. *arXiv preprint arXiv:2312.14238*.

Google DeepMind. 2024. Gemini 1.5 pro. https://deepmind.google/technologies/gemini/pro/.

Wenyi Hong, Weihan Wang, Ming Ding, Wenmeng Yu, Qingsong Lv, Yan Wang, Yean Cheng, Shiyu Huang, Junhui Ji, Zhao Xue, et al. 2024. Cogvlm2: Visual language models for image and video understanding. *arXiv preprint arXiv:2408.16500*.

Yuichi Inoue, Kento Sasaki, Yuma Ochi, Kazuki Fujii, Kotaro Tanahashi, and Yu Yamaguchi. 2024. Heron-bench: A benchmark for evaluating vision language models in japanese. In *CVPR workshop*.

Bo Li, Yuanhan Zhang, Dong Guo, Renrui Zhang, Feng Li, Hao Zhang, Kaichen Zhang, Yanwei Li, Ziwei Liu, and Chunyuan Li. 2024. Llava-onevision: Easy visual task transfer. *arXiv preprint arXiv:2408.03326*.

Fangyu Liu, Emanuele Bugliarello, Edoardo Maria Ponti, Siva Reddy, Nigel Collier, and Desmond Elliott. 2021. Visually grounded reasoning across languages and cultures. *arXiv preprint arXiv:2109.13238*.

Haotian Liu, Chunyuan Li, Yuheng Li, and Yong Jae Lee. 2024. Improved baselines with visual instruction tuning. In *CVPR*.

Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. 2023. Visual instruction tuning. In *NeurIPS*.

OpenAI. 2024. Gpt-4o.

Jonas Pfeiffer, Gregor Geigle, Aishwarya Kamath, Jan-Martin O Steitz, Stefan Roth, Ivan Vulić, and Iryna Gurevych. 2021. xgqa: Cross-lingual visual question answering. *arXiv preprint arXiv:2109.06082*.

Machel Reid, Nikolay Savinov, Denis Teplyashin, Dmitry Lepikhin, Timothy Lillicrap, Jean-baptiste Alayrac, et al. 2024. Gemini 1.5: Unlocking multimodal understanding across millions of tokens of context. *arXiv preprint arXiv:2403.05530*.

David Romero, Chenyang Lyu, Haryo Akbarianto Wibowo, Teresa Lynn, Injy Hamed, Aditya Nanda Kishore, Aishik Mandal, Alina Dragonetti, Artem Abzaliev, Atnafu Lambebo Tonja, et al. 2024. Cvqa: Culturally-diverse multilingual visual question answering benchmark. In *NeurIPS Datasets and Benchmarks Track*.

SakanaAI. 2024a. Ja-multi-image-vqa. https://huggingface.co/datasets/SakanaAI/JA-Multi-Image-VQA.

SakanaAI. 2024b. Ja-vg-vqa-500. https://huggingface.co/datasets/SakanaAI/JA-VG-VQA-500.

SakanaAI. 2024c. Ja-vlm-bench-in-the-wild. https://huggingface.co/datasets/SakanaAI/JA-VLM-Bench-In-the-Wild.

Jingqun Tang, Qi Liu, Yongjie Ye, Jinghui Lu, Shu Wei, Chunhui Lin, Wanqing Li, Mohamad Fitri Faiz Bin Mahmood, Hao Feng, Zhen Zhao, et al. 2024. Mtvqa: Benchmarking multilingual text-centric visual question answering. *arXiv preprint arXiv:2405.11985*.

Turing. 2024. Llava-bench-in-the-wild (japanese). https://github.com/turingmotors/heron/tree/main/playground/data/llava-bench-in-the-wild.

Le Xue, Manli Shu, Anas Awadalla, Jun Wang, An Yan, Senthil Purushwalkam, Honglu Zhou, Viraj Prabhu, Yutong Dai, Michael S Ryoo, et al. 2024. xgen-mm (blip-3): A family of open large multimodal models. *arXiv preprint arXiv:2408.08872*.

Xiang Yue, Yuansheng Ni, Kai Zhang, Tianyu Zheng, Ruoqi Liu, Ge Zhang, Samuel Stevens, Dongfu Jiang, Weiming Ren, Yuxuan Sun, et al. 2024a. Mmmu: A massive multi-discipline multimodal understanding and reasoning benchmark for expert agi. In *CVPR*.

Xiang Yue, Yueqi Song, Akari Asai, Seungone Kim, Jean de Dieu Nyandwi, Simran Khanuja, Anjali Kantharuban, Lintang Sutawika, Sathyanarayanan Ramamoorthy, and Graham Neubig. 2024b. Pangea: A

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fully open multilingual multimodal llm for 39 languages. *arXiv preprint arXiv:2410.16153*.

Xiang Yue, Tianyu Zheng, Yuansheng Ni, Yubo Wang, Kai Zhang, Shengbang Tong, Yuxuan Sun, Botao Yu, Ge Zhang, Huan Sun, Yu Su, Wenhu Chen, and Graham Neubig. 2024c. Mmmu-pro: A more robust multi-discipline multimodal understanding benchmark. *arXiv preprint arXiv:2409.02813*.

Kaichen Zhang, Bo Li, Peiyuan Zhang, Fanyi Pu, Joshua Adrian Cahyono, Kairui Hu, Shuai Liu, Yuanhan Zhang, Jingkang Yang, Chunyuan Li, et al. 2024. Lmms-eval: Reality check on the evaluation of large multimodal models. *arXiv preprint arXiv:2407.12772*.

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